Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 30 September 2026

Team ID: _____

Project Name: Script-Controlled ACL – Restrict Record Access Based on Field Value

Maximum Marks: 4 Marks

Technical Architecture:

The solution is implemented in ServiceNow using a script-controlled Access Control List (ACL). When a user requests a protected record, the ServiceNow security layer evaluates the applicable ACL. The ACL script checks the configured record field value and user criteria and grants or denies access accordingly.

Architecture flow: User → ServiceNow UI → ACL Security Layer → Script Evaluation → Field Value / User Criteria Validation → Allow or Deny Record Access → ServiceNow Database.

The architecture is primarily platform-based within the ServiceNow environment. No external API or machine-learning model is required for the core ACL implementation.

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	ServiceNow interface through which users request or view records.	ServiceNow UI / Platform UI
2	Application Logic-1	Script-controlled ACL evaluates whether the current user is permitted to access the record.	ServiceNow ACL + JavaScript

3	Application Logic-2	Condition and field-value validation used to determine record-level access.	ACL Condition / Script
4	Application Logic-3	User and role validation before allowing access to protected records.	ServiceNow User/Role APIs + ACL scripting
5	Database	Stores the target records, field values, user information, and configuration data.	ServiceNow Database
6	Cloud Database	ServiceNow-managed platform data storage for the application and its records.	ServiceNow Platform Data Layer
7	File Storage	Not required for the core ACL implementation.	Not Applicable
8	External API-1	No external API is required for field-based ACL evaluation.	Not Applicable
9	External API-2	No external API is required for user access validation in the core solution.	Not Applicable
10	Machine Learning Model	No machine-learning model is required for this rule-based access-control solution.	Not Applicable
11	Infrastructure (Server / Cloud)	Application runs on the ServiceNow cloud platform and uses the platform's managed infrastructure.	ServiceNow Cloud Platform

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The core solution does not require an external open-source framework. ACL scripting uses ServiceNow's supported server-side JavaScript capabilities.	ServiceNow JavaScript / ACL Framework
2	Security	Record access is controlled through ACL rules, field-value	ServiceNow ACL, Roles, Access Controls

		validation, user/role checks, and deny-by-default behavior where configured.	
3	Scalable Architecture	The solution can be extended by adding ACL rules, fields, roles, or reusable script logic without redesigning the complete application.	ServiceNow Platform / Modular ACL Design
4	Availability	The access-control logic runs within the ServiceNow platform whenever protected records are accessed.	ServiceNow Cloud Infrastructure
5	Performance	ACL scripts should use simple field and user checks and avoid unnecessary database operations to minimize access-evaluation overhead.	Optimized ServiceNow ACL Scripts

Architecture Components

1. User Layer – User attempts to access a protected ServiceNow record.
2. ServiceNow UI Layer – Receives the user's record request.
3. ACL Security Layer – Identifies and evaluates the applicable table/record ACL.
4. Script Evaluation Layer – Checks the configured field value and user/role criteria.
5. Decision Layer – Returns an allow or deny decision.
6. Data Layer – Provides the requested record from the ServiceNow database when access is permitted.

References

ServiceNow ACL documentation and platform security guidance can be used for implementation-specific configuration and validation.